

Existence and Uniqueness for the Sub-Riemannian Cubic Rhombus in Dimension 4 (simply connected, BV, free boundary conditions): Variational Proof

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Release 1.0
Manuscript CR-106-041

Abstract

For $n = 4$ a cubic rhombus $R_n(c)$ exists exactly for c in an explicit open interval, and is unique up to orthogonal maps. This manuscript states and proves exactly this proposition and nothing more. The title names a sub-Riemannian geometry, a simply connected topology, BV regularity, free boundary conditions and a variational technique. None of these enters the argument (Section 4). The proposition belongs to class S01, which 1490 manuscripts of this release share. This manuscript does not establish the result its title suggests. This manuscript does not support the hypothesis that it is an independent mathematical contribution.

1 Introduction

The cubic rhombus $R_n(c)$ is the parallelotope with unit edges and constant pairwise edge inner product c . Its Gram matrix is $(1 - c)I + cJ$, and every quantity below is computed from this one matrix. The classical background is the Gram determinant [1] and Hadamard's inequality [2]; see also [3]. This manuscript proves one proposition about this family. The title of the manuscript promises more than the proposition delivers. This manuscript does not support the hypothesis that it is an independent mathematical contribution.

2 Setting

Definition 1 (Cubic rhombus). For an integer $n \geq 2$ and $c \in \mathbb{R}$ let $G_n(c) = (1 - c)I_n + cJ_n$, where J_n is the $n \times n$ matrix of ones. A *cubic rhombus* $R_n(c)$ is the parallelotope $\{\sum_i t_i v_i : 0 \leq t_i \leq 1\} \subset \mathbb{R}^n$ spanned by unit vectors $v_1, \dots, v_n$ with $\langle v_i, v_j \rangle = c$ for $i \neq j$, that is, with Gram matrix $G_n(c)$. The case $c = 0$ is the unit cube, $n = 2$ gives a rhombus, and $n = 3$ gives a rhombohedron. The name “cubic rhombus” is used in this corpus for this family. It is not claimed to be standard terminology.

Lemma 1 (Spectrum and volume). *The matrix $G_n(c)$ has eigenvalue $1 + (n - 1)c$ with eigenvector $\mathbf{1} = (1, \dots, 1)^\top$, and eigenvalue $1 - c$ with multiplicity $n - 1$ on $\mathbf{1}^\perp$. Hence $\det G_n(c) = (1 - c)^{n-1}(1 + (n - 1)c)$, the matrix is positive definite if and only if $-\frac{1}{n-1} < c < 1$, and in that case $\text{vol}_n R_n(c) = \sqrt{\det G_n(c)}$.*

Proof. $J\mathbf{1} = n\mathbf{1}$, so $G\mathbf{1} = (1 - c + nc)\mathbf{1} = (1 + (n - 1)c)\mathbf{1}$. If $x \perp \mathbf{1}$ then $Jx = 0$, so $Gx = (1 - c)x$. These eigenspaces span $\mathbb{R}^n$, which gives the spectrum and the determinant, and positivity of both eigenvalues is the stated range. If V is the matrix with columns v_i then $G = V^T V$ and $\text{vol}_n = |\det V| = \sqrt{\det G}$. $\square$

Lemma 2 (Facets). *A facet of $R_n(c)$ is spanned by $n - 1$ of the edge vectors, and its Gram matrix is the principal $(n - 1) \times (n - 1)$ submatrix of $G_n(c)$, which is $G_{n-1}(c)$. Its $(n - 1)$ -volume is $\sqrt{(1 - c)^{n-2}(1 + (n - 2)c)}$, and the surface area of $R_n(c)$ is $2n$ times this number.*

Proof. Deleting a row and the matching column of $(1 - c)I + cJ$ leaves $(1 - c)I + cJ$ of order $n - 1$. Apply Lemma 1 in dimension $n - 1$. There are $2n$ facets, in n parallel pairs. $\square$

Lemma 3 (Logarithmic derivative). *On $(-\frac{1}{n-1}, 1)$ let $F(c) = \ln \det G_n(c) = (n - 1) \ln(1 - c) + \ln(1 + (n - 1)c)$. Then*

$$F'(c) = -\frac{n(n - 1)c}{(1 - c)(1 + (n - 1)c)},$$

the denominator is positive, and so F' has the sign of $-c$.

Proof. $F'(c) = -\frac{n-1}{1-c} + \frac{n-1}{1+(n-1)c} = (n - 1) \frac{(1-c)-(1+(n-1)c)}{(1-c)(1+(n-1)c)} = -\frac{n(n-1)c}{(1-c)(1+(n-1)c)}$. $\square$

Lemma 4 (Vertex Gram matrices). *The vertices of $R_n(c)$ are the points $\sum_{i \in S} v_i$ for $S \subseteq \{1, \dots, n\}$. At the vertex S the unit edge directions are $\varepsilon_i v_i$ with $\varepsilon_i = -1$ for $i \in S$ and $\varepsilon_i = +1$ otherwise, so their Gram matrix is $D_\varepsilon G_n(c) D_\varepsilon$ with $D_\varepsilon = \text{diag}(\varepsilon)$. Suppose $n \geq 3$ and that all off-diagonal entries of $D_\varepsilon G_n(c) D_\varepsilon$ equal one number s . Then $s = c$. For $n = 2$ the conclusion fails: the vertices with $\varepsilon_1 \varepsilon_2 = -1$ give $s = -c$.*

Proof. The first statements are immediate from the definition. The off-diagonal entries are $\varepsilon_i \varepsilon_j c$. If $c = 0$ then $s = 0 = c$. If $c \neq 0$ put $p = s/c$, so $\varepsilon_i \varepsilon_j = p$ for all $i \neq j$. For distinct i, j, k we have $\varepsilon_i \varepsilon_j \cdot \varepsilon_i \varepsilon_k = \varepsilon_j \varepsilon_k$, that is $p^2 = p$, so $p = 1$. $\square$

3 Result

Theorem 1 (Existence and uniqueness). *Let $n = 4$. Then $\det G_n(c) = (1 - c)^3(3c + 1)$ and $G_n(c)$ is positive definite if and only if $-\frac{1}{3} < c < 1$. For every such c there is a cubic rhombus $R_n(c)$, and any two cubic rhombi with the same (n, c) differ by an orthogonal transformation of $\mathbb{R}^n$. If $c < -\frac{1}{3}$ or $c > 1$ there is no system of n unit vectors with pairwise inner product c .*

Proof. The determinant and the range of positivity are Lemma 1. If $G_n(c)$ is positive definite it has a Cholesky factorization $G_n(c) = V^T V$ with V invertible, and the columns of V are unit vectors with the required inner products. This gives existence. If W is another matrix with $W^T W = G_n(c)$, put $Q = WV^{-1}$. Then $Q^T Q = V^{-T} G_n(c) V^{-1} = I$, so Q is orthogonal and $W = QV$. This gives uniqueness. Finally, the Gram matrix of any system of vectors is positive semidefinite, so a system exists only if both eigenvalues in Lemma 1 are nonnegative, that is, only if $-\frac{1}{3} \leq c \leq 1$. $\square$

4 Parameters carried in the title

Parameter	Status	Role in the argument
Dimension	Operative	The proposition is stated for dimension 4.
Contribution	Operative	The label corresponds to the proposition proved here.
Geometry: sub-Riemannian	Not used	Every statement concerns the Euclidean metric on $\mathbb{R}^n$.
Topology: simply connected	Not used	$R_n(c)$ is a convex polytope, hence contractible. No other topology is considered.
Regularity: BV	Not used	The object is a polytope. No function space on it is considered, and the label is carried as metadata.
Boundary conditions: free	Not used	No boundary value problem is posed. The word boundary in Theorem 1 refers to the facets of the polytope, where it occurs.
Technique: variational	Not used	The proof given is by linear algebra (eigenvalues and a Cholesky factorization). The label is carried as metadata.

Of the seven parameters of the title, two enter the argument and five do not. The five that do not are retained so that the title is complete.

5 Relation to the rest of the release

The proposition of this manuscript is in class S01 (scope: general n). In this release, 1490 manuscripts carry class S01. The principal member of this class in this release is manuscript CR-001-001. This manuscript restates that proposition under different title parameters. Counting manuscripts is therefore not counting results. This manuscript does not support the hypothesis that it is an independent mathematical contribution.

6 Verification status

Item	Status
Human review	none
Independent scrutiny	none
Lean formalization	not attempted
Exact computer algebra checks tagged to this class	6 of 6 passed
Novelty of the proposition	not assessed

The computer algebra checks confirm closed forms against direct computation for $n = 2, \dots, 8$ and confirm the symbolic identities in n . They do not address novelty or significance. Some results could have issues, and a correction would be recorded as a new version.

7 Statement of non-support

This manuscript proves a proposition about a determinant of a matrix of the form $(1 - c)I + cJ$. It does not prove, support, or provide evidence for the claim that the proposition is an independent

result, a new result, or a result of the kind the title suggests. This manuscript does not support the hypothesis that it is an independent mathematical contribution.

References

- [1] J. P. Gram, *Über die Entwicklung reeller Functionen in Reihen mittelst der Methode der kleinsten Quadrate*, J. Reine Angew. Math. 94 (1883), 41–73.
- [2] J. Hadamard, *Résolution d’une question relative aux déterminants*, Bull. Sci. Math. 17 (1893), 240–246.
- [3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.